

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – PROMPT ENGINEERING CHALLENGE

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL2, BL3)	Assessment:	Assessment Tool 1 – Transformer Mechanics Worksheet
Weightage:	40% of CIA	Total Marks:	2 * 50 = 100 (2 Questions)
CO1 Statement:	<i>pply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems (BL2, BL3)</i>		
Student Name:	B. Lakshmi Sagar	Reg. No.:	192472070
Section / Batch:	B.E –CSE-AI	Date:	28/07/2026

Instructions to Students

- Answer 2 questionss.Each questions carries 50 marks..Total100 Marks.
1. Design appropriate prompts for the given tasks using suitable prompt engineering techniques.
 2. Select and apply the appropriate prompting strategy (Zero-shot, One-shot, Few-shot, Chain-of-Thought, or Structured Prompting) based on the given scenario.
 3. Include clear instructions, relevant context, necessary constraints, and the expected output format while designing each prompt.
 4. Evaluate and refine the designed prompts to improve their clarity, effectiveness, and quality by applying prompt engineering best practices.
 5. Design prompts for a variety of real-world applications, such as text generation, summarization, translation, question answering, code generation, and conversational AI.
 6. Justify the designed prompt by briefly explaining the selected prompting technique and how it helps achieve the desired output.

Code of Conduct

I LAKSHMI SAGAR(192472070) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Lakhmi sagar

SECTION A – PROMPT ENGINEERING CHALLENGE

Question 12: Travel Itinerary Prompt (Chennai, 3 Days)

Prompting Technique Used: Structured Prompting

Structured prompting fits best here because a travel itinerary has multiple distinct components (day-wise breakdown, timings, categories like food, sightseeing, transport) that are much easier for the model to generate correctly when the prompt itself is broken into labeled sections, rather than one block of instructions.

The Prompt:

You are an experienced local travel guide specializing in Chennai, India.

Task: Create a detailed 3-day travel itinerary for a first-time tourist visiting Chennai.

Context:

- Traveler type: A couple in their late 20s, moderate budget, interested in culture, temples, beaches, and local food.
- Trip duration: 3 days, 2 nights
- Starting point: Chennai Central Railway Station
- Mode of travel within city: Cab/Auto

Constraints:

- Include a mix of morning, afternoon, and evening activities for each day.
- Keep each day's plan realistic (no more than 4-5 activities per day).
- Include at least one authentic local food recommendation per day.
- Mention approximate time required for each activity.
- Keep the budget moderate (avoid only high-end luxury suggestions).

Expected Output Format:

Day 1:

- Morning: [Activity + short description + time required]
- Afternoon: [Activity + short description + time required]
- Evening: [Activity + short description + time required]
- Food Recommendation: [Dish/restaurant]

(Repeat the same format for Day 2 and Day 3)

At the end, add a short "Travel Tips" section with 3 practical tips for getting around Chennai.

Justification:

The structured format (Context, Constraints, Output Format) gives the LLM clear boundaries, preventing generic or overly long responses. Labeling the output format explicitly (Morning/Afternoon/Evening/Food) ensures consistency across all 3 days and makes the output directly usable, which is the core goal of structured prompting — precision and predictability of format (aligning with CO2: Precision).

Question 39: AI Career Guidance Assistant Prompt

Prompting Technique Used: Few-Shot Prompting

Few-shot prompting is ideal here because career recommendation is a pattern-matching task — showing the model 2-3 examples of how "interest to recommended career path" mappings should look ensures consistent reasoning style and output format across many different student inputs, which zero-shot alone often fails to standardize.

The Prompt:

You are an AI Career Guidance Assistant that helps students discover suitable career paths based on their interests, strengths, and preferred subjects.

Task: Given a student's interests, recommend 2-3 suitable career paths, along

with a brief explanation and the skills/subjects they should focus on.

Follow the pattern shown in these examples:

Example 1:

Student Interest: "I enjoy solving puzzles, coding, and building small apps."

Output:

1. Software Developer - Strong problem-solving and coding interest align well with this role. Focus on: Data Structures, Algorithms, Programming Languages.
2. Data Scientist - Enjoying logical puzzles suggests aptitude for analytical thinking. Focus on: Statistics, Python, Machine Learning.

Example 2:

Student Interest: "I like biology, helping people, and watching medical shows."

Output:

1. Doctor (MBBS) - Interest in biology and helping people is a strong indicator. Focus on: Biology, Chemistry, NEET preparation.
2. Biomedical Engineer - Combines biology interest with technical problem-solving. Focus on: Biology, Physics, Engineering fundamentals.

Now generate a similar response for this student:

Student Interest: "{insert student's interest here}"

Constraints:

- Recommend exactly 2-3 career paths (not more, not less).
- Each recommendation must include: Career name, 1-line reasoning, and subjects/skills to focus on.
- Keep tone encouraging and age-appropriate for a high school/undergrad student.
- Avoid recommending careers unrelated to the stated interests.

Output Format:

1. [Career Name] - [Reasoning]. Focus on: [Skills/Subjects]
 2. [Career Name] - [Reasoning]. Focus on: [Skills/Subjects]
- (optional 3rd)

Justification:

Few-shot examples "teach" the model the exact reasoning pattern and formatting style expected, which is critical for a career guidance tool that will be used repeatedly with different student inputs — it needs to behave consistently. This reduces ambiguity and hallucination risk compared to zero-shot, directly supporting CO2's precision requirement (BL2, BL3: applying prompt engineering techniques to solve a real-world problem).

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technique Selection	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Output Effectiveness	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Prompt Craftsmanship	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Prompt-Output Documentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Design Rationale	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____